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(57) Abstract:

ABSTRACT CONTEXT-ADAPTIVE EXPLAINABLE ARTIFICIAL INTELLIGENCE SYSTEM FOR BRAND-LOGO TRUST-PROPENSITY INFERENCE AND AUTOMATED DESIGN PRESCRIPTION A method for context-adaptive brand-logo trust-propensity inference includes extracting multi-domain features (130) from a logo image, mapping a colour vector to emotion-labelled palettes (150), generating a trust-propensity coefficient (160), generating raw feature attributions (170), classifying a chromatic operating condition and producing a context-calibrated attribution vector while preserving the raw attribution vector at a Dynamic Signal Calibration Module (180), generating a quantitative design prescription (190), and outputting the trust-propensity score and design prescription (210).

FORM-2

**THE PATENTS ACT, 1970
(39 of 1970)**

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THE PATENTS RULES, 2003

**COMPLETE
SPECIFICATION**

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(See section 10 and rule 13)

TITLE OF INVENTION

15 **CONTEXT-ADAPTIVE EXPLAINABLE ARTIFICIAL INTELLIGENCE
SYSTEM FOR BRAND-LOGO TRUST-PROPENSITY INFERENCE AND
AUTOMATED DESIGN PRESCRIPTION**

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**THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES
THE INVENTION AND THE MANNER IN WHICH IT IS TO BE
PERFORMED**

CROSS-REFERENCE TO RELATED APPLICATIONS AND PRIORITY

[0001] The present application does not claim priority from any patent application.

PREAMBLE

5 [0002] The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

[0003] The present disclosure relates to explainable artificial intelligence systems for computer vision and brand-identity analytics, and more particularly to
10 a context-adaptive system for inferring a trust-propensity coefficient from a brand-
logo image, generating chromatic-condition-dependent feature attributions, and
producing quantitative design prescriptions.

BACKGROUND

[0004] The following text of background art is provided for purposes of
15 understanding and is not an admission that any cited document or technique forms
part of the prior art under applicable law.

[0005] Brand logos communicate identity through interacting chromatic,
geometric and typographic attributes. Existing logo-processing systems primarily
recognise, retrieve, classify or generate logos. Design evaluation remains dependent
20 on human opinion, focus groups or general-purpose feature-importance tools that
apply the same interpretation logic to every image. Uniform analysis creates
misleading output in at least two situations: a monochromatic logo may receive an
irrelevant colour recommendation, and an intentionally dominant brand colour may
be treated as a defect merely because its raw model attribution is negative.
25 Conventional explainability also identifies influential features without determining
a bounded, actionable modification capable of improving the predicted outcome.

[0006] US10769496B2, titled "Logo Detection" and assigned to Adobe Inc.,
discloses a system that receives an image, detects candidate logo regions, extracts
neural feature vectors and compares them with reference-logo vectors to identify a
30 logo. This reference addresses logo recognition and localisation but does not infer

a trust-propensity coefficient, map a logo to an emotion manifold, adapt explanations to chromatic conditions, or generate a quantitative design prescription.

[0007] US10558758B2, titled "Enhancing a Computer to Match Emotion and Tone in Text with the Emotion and Tone Depicted by the Color in the Theme of the Page or Its Background" and assigned to International Business Machines Corporation, associates emotion or tone derived from text with page-theme or background colour. This reference processes text and page colour rather than logo-image geometry, typography and learned visual representations. The reference does not perform matched-pair logo analysis, logo trust inference or explainability-driven redesign.

[0008] US10467526B1, titled "Artificial Intelligence System for Image Similarity Analysis Using Optimized Image Pair Selection and Multi-Scale Convolutional Neural Networks" and assigned to Amazon Technologies, Inc., uses selected image pairs and multi-scale convolutional neural networks for image-similarity analysis. This reference measures general image similarity but does not use a controlled baseline/revision logo pair to isolate a design intervention, does not derive a psychographic trust target, and does not generate context-calibrated attributions or logo-design prescriptions.

[0009] Peer-reviewed literature has established that colour affects trust judgements, that logo naturalness and colour influence brand-gender perception and affective response, and that perceived logo colourfulness influences consumer judgements through perceived product variety and brand positioning. However, these studies provide consumer-behaviour evidence rather than an automated logo-analysis architecture. None of the reviewed references discloses matched-pair differential inference, adaptive explanation routing based on chromatic conditions, or quantitative design prescription.

[0010] The present disclosure addresses the above shortcomings of the prior art. However, the subject matter disclosed herein is entirely different from the prior art in terms of novelty and technological advancements.

SUMMARY

[0011] The following summary is intended to introduce aspects of the present invention in a simplified form and does not identify all the features or the scope of the claimed invention. The summary is provided to assist in understanding the invention and is to be read in conjunction with the detailed description and accompanying drawings, claims, and abstract. One or more embodiments of the present disclosure are set forth below solely for illustrative purposes and should not be construed as limiting.

[0012] According to an aspect of the present disclosure, a method for context-adaptive brand-logo trust-propensity inference and design prescription is provided.

10 The method includes receiving and validating a logo image or a baseline/revision pair at an Image Input and Validation Module. The method includes preprocessing the logo image and extracting multi-domain features including chromatic features, geometric features, typographic features and learned image embeddings at an Image Preprocessing Module and a Multi-Domain Feature Extraction Module. The method includes computing differential features when a matched pair is supplied at a Matched-Pair Differential Encoder. The method includes mapping a representative colour vector to neighbouring emotion-labelled palette vectors at an Emotion-Manifold Mapping Module. The method includes generating a trust-propensity coefficient at a Trust-Inference Engine. The method includes generating raw feature attributions at an Explainability Engine. The method includes classifying a chromatic operating condition and generating a context-calibrated attribution vector while preserving the raw attribution vector at a Dynamic Signal Calibration Module. The method includes selecting a modifiable feature and generating a quantitative design prescription at a Design-Prescription Generator.

25 The method includes outputting a trust-propensity score, explanations and design prescription at a Reporting and Visualisation Module.

[0013] The above aspect provides a technical advantage of enabling automated, auditable trust-propensity inference from brand-logo images while adapting explanation logic to chromatic operating conditions, thereby reducing irrelevant colour-based recommendations for achromatic logos and separating intentional dominant-colour strategy from potentially modifiable design friction.

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[0014] According to other aspects of the present disclosure, the method may include one or more of the following features. The chromatic operating condition may be classified as achromatic, ordinarily chromatic, or characterised by a dominant visual signal. The Dynamic Signal Calibration Module may include an
5 Achromatic Signal Gate that excludes colour-domain attribution values from a prescription pathway when measured saturation is below a configurable threshold.

[0015] The above aspect provides a technical advantage of preventing colour features from controlling design recommendations when colour is non-operative in the logo, thereby avoiding irrelevant palette advice for monochromatic or near-
10 monochromatic logos.

[0016] According to other aspects of the present disclosure, the Dynamic Signal Calibration Module may include a Dominant Signal Contextualiser that retains the raw attribution and generates a contextual preservation/modification score when a dominance ratio exceeds a configurable threshold. The method may
15 include storing both an unmodified raw attribution vector and a distinct context-calibrated attribution vector along with thresholds, model version and selected analysis mode at an Audit and Provenance Store.

[0017] The above aspect provides a technical advantage of preserving explanation fidelity through independent raw and calibrated attribution streams,
20 enabling intentional dominant brand colours to be recognised and handled appropriately rather than being flagged as defects.

[0018] According to other aspects of the present disclosure, the Matched-Pair Differential Encoder may compute an embedding difference between baseline and revised logo embeddings and may compute an embedding displacement as a
25 displacement feature. The Matched-Pair Differential Encoder may retain engineered feature differences to support feature-specific interpretation of a design intervention.

[0019] The above aspect provides a technical advantage of enabling reproducible measurement of a controlled logo change rather than comparison of
30 unrelated images, thereby isolating the effect of deliberate design interventions on trust-propensity.

[0020] According to other aspects of the present disclosure, the Emotion-Manifold Mapping Module may implement nearest-neighbour search through a KD-tree or equivalent indexing structure. The Emotion-Manifold Mapping Module may use a corpus of palettes tagged across psychological dimensions. The Trust-Inference Engine may use a gradient-boosted decision-tree model to generate the normalised trust-propensity coefficient.

[0021] The above aspect provides a technical advantage of grounding the predictive target in an independently prepared emotion manifold, reducing circular dependence on labels generated from the same engineered features used for model training.

[0022] According to other aspects of the present disclosure, the method may include combining outputs from a differential model and a general-logo model using configurable weights at an Ensemble Corroboration Module. The Design-Prescription Generator may select a modifiable feature with an adverse calibrated contribution and compute a bounded change including reduction of stroke weight, increase of inter-character spacing, adjustment of circularity, reduction of edge-density imbalance, or constrained palette adjustment.

[0023] The above aspect provides a technical advantage of improving robustness through cross-model corroboration and converting model explanations into auditable, numerical design actions rather than general feature-importance rankings.

[0024] According to other aspects of the present disclosure, the method may include constructing and re-evaluating bounded logo variations at a Candidate Modification Module by evaluating candidate changes and selecting a minimum change satisfying an improvement criterion and applicable design-preservation constraints.

[0025] The above aspect provides a technical advantage of generating actionable bounded design modifications with minimum effective change selection, enabling designers to implement precise, constrained adjustments that achieve trust-propensity improvement while preserving brand identity.

[0026] The foregoing general description of the illustrative embodiments and the following detailed description thereof are merely exemplary aspects of the teachings of this disclosure and are not restrictive.

BRIEF DESCRIPTION OF FIGURES

5 [0027] The drawings appended herein illustrate exemplary embodiments of the present invention and are provided to enhance understanding of the inventive features disclosed. These figures, when read in conjunction with the detailed description, depict the operational flow, component architecture, and other mechanisms.

10 [0028] FIG. 1 illustrates an overall architecture of a Differential Trust Engine, according to aspects of the present disclosure.

[0029] FIG. 2 illustrates an end-to-end method performed by the Differential Trust Engine, according to aspects of the present disclosure.

15 [0030] FIG. 3 illustrates a Dynamic Signal Calibration Module producing a context-calibrated attribution vector, according to aspects of the present disclosure.

[0031] FIG. 4 illustrates controlled matched-pair differential analysis performed by a Matched-Pair Differential Encoder, according to aspects of the present disclosure.

20 [0032] FIG. 5 illustrates a counterfactual design-prescription workflow, according to aspects of the present disclosure.

DETAILED DESCRIPTION

25 [0033] The following detailed description is intended to provide an in-depth understanding of the invention and its various components, configurations, and operational aspects. While specific embodiments are described with reference to the accompanying drawings, the invention is not limited to the particular forms disclosed herein. The scope of the invention shall be interpreted broadly and in accordance with the claims appended hereto.

30 [0034] The terms “comprises,” “comprising,” “includes,” “including,” and similar expressions used in this description are intended to be open-ended and should be interpreted to include, but not be limited to, the mentioned elements.

Unless explicitly stated otherwise, singular terms may include their plural equivalents and vice versa, depending on the context in which they appear.

[0035] Reference is now made to the accompanying figures, which illustrate example embodiments of the present invention. It should be appreciated that the figures are intended for illustration and explanatory purposes only and do not limit the scope of the invention. Similar reference numerals have been used to denote functionally similar components throughout the figures.

[0036] The embodiments described herein are presented for illustrative purposes and are subject to variations, modifications, and adaptations by those skilled in the art. Any equivalent implementations that perform substantially the same function in substantially the same manner are intended to fall within the scope of this disclosure and the claims appended hereto.

[0037] The use of the term “exemplary” in the description shall be understood to mean “serving as an example or illustration” and shall not be construed as limiting the invention to the preferred embodiments disclosed. Various other embodiments may be developed without departing from the spirit or essential characteristics of the invention. The technical principles and advantages outlined in the foregoing description should be read in conjunction with the figures and claims. It is expressly understood that while the detailed description sets out specific structural and operational aspects, the scope of protection is governed by the claims and includes all legal equivalents of the claimed subject matter. Embodiments of the disclosure are described in the following paragraphs with reference to Figures. In Figures, the same elements or elements that have the same functions are indicated by the same reference signs.

[0038] The present invention will now be described in detail with reference to the accompanying drawings illustrating exemplary embodiments.

[0039] FIG. 1 illustrates an overall architecture of a Differential Trust Engine 100 according to various embodiments. The Differential Trust Engine 100 may operate in two distinct input modes: a single-logo mode where a single logo image is analyzed independently, and a matched-pair mode where a baseline logo r0 and

a controlled revision r_1 are processed together to isolate the effect of a deliberate design intervention.

[0040] An Image Input and Validation Module 110 may receive either a single logo image or a matched baseline/revision pair. The Image Input and Validation
5 Module 110 may verify file type, dimensions, transparency and integrity of the input. In some cases, the Image Input and Validation Module 110 validates that the received logo image conforms to supported formats and resolution requirements before passing the validated input to subsequent processing stages.

[0041] An Image Preprocessing Module 120 may perform resizing,
10 normalisation, colour-space conversion, background handling and optional foreground segmentation on the received logo images. The Image Preprocessing Module 120 may convert the logo image to a standardised representation suitable for feature extraction.

[0042] A Multi-Domain Feature Extraction Module 130 may extract chromatic
15 features including colour statistics, saturation and dominant-channel ratio from the logo image. The Multi-Domain Feature Extraction Module 130 may extract geometric features including contrast, circularity, symmetry, stroke characteristics, edge density and spatial balance from the logo image. The Multi-Domain Feature Extraction Module 130 may extract typography-related descriptors from the logo
20 image. The Multi-Domain Feature Extraction Module 130 may extract a learned image embedding using a neural network encoder from the logo image. In some cases, the Multi-Domain Feature Extraction Module 130 may use ResNet50 as an exemplary neural network encoder for learned image embeddings.

[0043] A Matched-Pair Differential Encoder 140 may compute differential
25 features when a matched pair is supplied. The Matched-Pair Differential Encoder 140 may compute an embedding difference $\Delta E = E(r_1) - E(r_0)$ between baseline and revised logo embeddings. The Matched-Pair Differential Encoder 140 may compute an embedding displacement $d = \|\Delta E\|_2$ as a displacement feature between the baseline and revised logos. The Matched-Pair Differential Encoder 140 may
30 retain engineered feature differences Δf to support feature-specific interpretation of

the design intervention, capturing typography, stroke weight, spacing, geometry or colour modifications.

[0044] An Emotion-Manifold Mapping Module 150 may map a dominant or representative colour vector c to k neighbouring palette vectors p_j using nearest-neighbour search. The Emotion-Manifold Mapping Module 150 may calculate an affective target as a distance-weighted combination of emotion dimensions associated with neighbouring palettes. The Emotion-Manifold Mapping Module 150 may implement nearest-neighbour search through a KD-tree or equivalent indexing structure. In some cases, the Emotion-Manifold Mapping Module 150 may use a corpus of 745 palettes tagged across 35 psychological dimensions, expanded to 3,725 palette-associated records.

[0045] A Trust-Inference Engine 160 may generate a normalised trust-propensity coefficient using a gradient-boosted decision-tree model. In some cases, the Trust-Inference Engine 160 may use LightGBM as an exemplary gradient-boosted decision-tree implementation. The Trust-Inference Engine 160 receives the extracted features and mapped affective target to produce a preliminary trust-propensity coefficient.

[0046] An Ensemble Corroboration Module 165 may combine outputs from a differential model T_d and a general-logo model T_g using the formula $T_f = \alpha T_d + (1-\alpha)T_g$ with configurable weights. In some cases, the Ensemble Corroboration Module 165 may use an illustrative laboratory configuration with $\alpha = 0.60$ for combining differential and general-logo model outputs.

[0047] An Explainability Engine 170 may generate raw local feature-attribution values for the current prediction using model-agnostic explanation techniques. In some cases, the Explainability Engine 170 may use SHAP as an exemplary technique for generating raw local feature-attribution values.

[0048] A Dynamic Signal Calibration Module 180 may classify the chromatic operating condition as achromatic, ordinarily chromatic, or characterised by a dominant visual signal. The Dynamic Signal Calibration Module 180 may comprise an Achromatic Signal Gate 181 and a Dominant Signal Contextualiser 182. The Dynamic Signal Calibration Module 180 may maintain dual attribution records

storing both an unmodified raw attribution vector and a distinct context-calibrated attribution vector along with thresholds, model version and selected analysis mode.

[0049] A Design-Prescription Generator 190 may select a modifiable feature with an adverse calibrated contribution and compute a bounded change. The
5 bounded change may include reduction of stroke weight, increase of inter-character spacing, adjustment of circularity, reduction of edge-density imbalance, or constrained palette adjustment.

[0050] A Candidate Modification Module 200 may construct and re-evaluate bounded logo variations. The Candidate Modification Module 200 may evaluate
10 candidate changes Δx and select the minimum change satisfying the improvement criterion $\hat{T}(x+\Delta x) - \hat{T}(x) \geq \delta$ and applicable design-preservation constraints.

[0051] A Reporting and Visualisation Module 210 may produce output including the trust-propensity score, active analysis mode, raw and calibrated explanations, design prescription and before/after comparison report.

[0052] An Audit and Provenance Store 220 may retain input metadata, extracted features, raw and calibrated attributions, thresholds, model version, selected mode and output history for auditability. The Audit and Provenance Store
15 220 receives data from the Dynamic Signal Calibration Module 180, the Design-Prescription Generator 190, the Candidate Modification Module 200 and the
20 Reporting and Visualisation Module 210.

[0053] The Differential Trust Engine 100 may be implemented as an API, design-audit application or computer-assisted logo-design module. The modular architecture enables integration with existing brand-identity workflows and design tools.

[0054] FIG. 2 illustrates an end-to-end method performed by the Differential Trust Engine 100 according to various embodiments. The method maintains separation between raw and calibrated attribution streams throughout the process, with each step contributing to the final auditable output.

[0055] The method may begin with logo reception and validation at the Image
30 Input and Validation Module 110. The Image Input and Validation Module 110 may receive and validate a logo image or a baseline/revision pair, verifying file

type, dimensions, transparency and integrity before passing the validated input to subsequent processing stages.

[0056] The method may proceed with preprocessing at the Image Preprocessing Module 120. The Image Preprocessing Module 120 may perform
5 resizing, normalisation, colour-space conversion, background handling and optional foreground segmentation on the received logo images.

[0057] The method may proceed with multi-domain feature extraction at the Multi-Domain Feature Extraction Module 130. The Multi-Domain Feature Extraction Module 130 may extract chromatic features, geometric features,
10 typographic features and learned image embeddings from the preprocessed logo image.

[0058] When a matched pair is supplied, the method may proceed with differential feature computation at the Matched-Pair Differential Encoder 140. The Matched-Pair Differential Encoder 140 may compute the embedding difference ΔE
15 $= E(r1) - E(r0)$, the displacement magnitude $d = \|\Delta E\|_2$, and engineered feature differences Δf between the baseline and revised logos.

[0059] The method may proceed with colour-to-emotion mapping at the Emotion-Manifold Mapping Module 150. The Emotion-Manifold Mapping Module 150 may map a representative colour vector to neighbouring emotion-
20 labelled palette vectors and calculate a distance-weighted affective target.

[0060] The method may proceed with trust-propensity generation at the Trust-Inference Engine 160. The Trust-Inference Engine 160 may generate a normalised trust-propensity coefficient from the extracted features and mapped affective target.

[0061] The method may optionally proceed with ensembling at the Ensemble Corroboration Module 165. The Ensemble Corroboration Module 165 may
25 combine outputs from a differential model T_d and a general-logo model T_g using the formula $T_f = \alpha T_d + (1-\alpha)T_g$ with configurable weights.

[0062] The method may proceed with raw attribution retention at the Explainability Engine 170. The Explainability Engine 170 may generate raw local
30 feature-attribution values for the current prediction and retain the raw attribution vector for subsequent processing and audit purposes.

[0063] The method may proceed with chromatic condition classification and context-calibrated attribution generation at the Dynamic Signal Calibration Module 180. The Dynamic Signal Calibration Module 180 may classify the chromatic operating condition and produce a context-calibrated attribution vector without
5 overwriting the raw attribution vector. The Achromatic Signal Gate 181 may activate when measured saturation S is below the achromatic threshold τ_a to exclude colour-domain entries from the prescription pathway. The Dominant Signal Contextualiser 182 may activate when a dominance ratio D exceeds the dominance threshold τ_d to retain the raw attribution and generate a contextual
10 preservation/modification score.

[0064] The method may proceed with modifiable feature selection and quantitative prescription generation at the Design-Prescription Generator 190. The Design-Prescription Generator 190 may select a modifiable feature with an adverse calibrated contribution and compute a bounded design modification.

15 [0065] The method may optionally proceed with candidate logo construction and re-evaluation at the Candidate Modification Module 200. The Candidate Modification Module 200 may construct candidate logo variations, re-evaluate each candidate against the trust-propensity improvement criterion $\hat{T}(x+\Delta x) - \hat{T}(x) \geq \delta$ and design-preservation constraints, and select the minimum effective change
20 satisfying the threshold δ and all constraints.

[0066] The method may conclude with output of score, explanations, prescription and report through the Reporting and Visualisation Module 210. The Reporting and Visualisation Module 210 may produce the trust-propensity score, active analysis mode, raw and calibrated explanations, design prescription and
25 before/after comparison report. The audit trail may be written to the Audit and Provenance Store 220, which retains input metadata, extracted features, raw and calibrated attributions, thresholds, model version, selected mode and output history.

[0067] FIG. 3 illustrates the Dynamic Signal Calibration Module 180 producing a context-calibrated attribution vector from a raw attribution vector
30 generated by the Explainability Engine 170 according to various embodiments. The Dynamic Signal Calibration Module 180 evaluates measured saturation S and

dominance ratio D against configurable thresholds to determine the appropriate processing pathway.

[0068] The Achromatic Signal Gate 181 may activate when measured saturation S is below an achromatic threshold τ_a . In some cases, the Achromatic Signal Gate 181 may use an illustrative laboratory threshold of $\tau_a = 0.05$ for detecting non-operative colour conditions. When the Achromatic Signal Gate 181 activates, colour-domain entries may be set to zero only in the prescription vector Φ_{cal} while preserving the raw attribution vector Φ_{raw} . The Achromatic Signal Gate 181 excludes colour features from the prescription pathway, redirecting the diagnostic process to geometry, typography and structure while avoiding irrelevant palette advice for monochromatic or near-monochromatic logos.

[0069] When saturation S exceeds the achromatic threshold τ_a , the Dynamic Signal Calibration Module 180 may evaluate whether the dominance ratio D exceeds a dominance threshold τ_d . The Dominant Signal Contextualiser 182 may activate when the dominance ratio D exceeds the threshold τ_d . In some cases, the Dominant Signal Contextualiser 182 may use an illustrative dominance threshold of $\tau_d = 0.40$ for detecting dominant visual signals.

[0070] When the Dominant Signal Contextualiser 182 activates, the raw attribution vector Φ_{raw} may be retained and a contextual preservation/modification score ψ may be generated without altering the raw attribution. The Dominant Signal Contextualiser 182 may compute the contextual score $\psi = h(\Phi_{raw}, D, \text{model consistency, optional preservation constraints})$ to determine whether the dominant signal is preserved, cautiously modified, or modified. The contextual score enables intentional dominant brand colours to be recognised and handled appropriately rather than being flagged as defects.

[0071] When the dominance ratio D does not exceed the threshold τ_d and saturation S exceeds the achromatic threshold τ_a , the Dynamic Signal Calibration Module 180 may apply standard chromatic interpretation. In the standard chromatic interpretation pathway, the raw attribution values may be processed without achromatic gating or dominant-signal contextualisation.

[0072] The three conditional paths may converge to produce the calibrated attribution vector Φ_{cal} . The Audit and Provenance Store 220 may maintain dual attribution records storing both the unmodified raw attribution vector Φ_{raw} and the distinct context-calibrated attribution vector Φ_{cal} together with the thresholds τ_a and τ_d , model version and selected analysis mode for auditability. The dual storage architecture ensures that the unmodified raw attribution vector remains available for audit purposes while the separately computed calibrated attribution vector is used for generating design prescriptions.

[0073] FIG. 4 illustrates controlled matched-pair differential analysis performed by the Matched-Pair Differential Encoder 140 according to various embodiments. The matched-pair differential analysis enables the system to isolate and quantify the effect of deliberate design interventions by comparing a baseline logo against a controlled revision.

[0074] A baseline logo r_0 and a controlled revision r_1 may each be processed by a visual encoder to obtain respective embeddings $E(r_0)$ and $E(r_1)$. The visual encoder may extract learned representations from each logo image that capture visual characteristics relevant to trust-propensity inference.

[0075] Engineered feature differences Δf may be computed from the baseline logo r_0 and controlled revision r_1 to capture typography, stroke weight, spacing, geometry or colour modifications. The engineered feature differences Δf may be retained to support feature-specific interpretation of the design intervention, enabling identification of which specific design variables contribute to changes in trust-propensity.

[0076] The Matched-Pair Differential Encoder 140 may compute the feature difference $\Delta E = E(r_1) - E(r_0)$ between the baseline and revised logo embeddings. The embedding difference ΔE represents the change in the learned representation space resulting from the design intervention.

[0077] The Matched-Pair Differential Encoder 140 may compute an embedding displacement $d = \|\Delta E\|_2$ as a displacement feature between the baseline and revised logos. The displacement magnitude d quantifies the intervention

magnitude in the embedding space, providing a scalar measure of the overall change between the baseline and revised logos.

[0078] The Matched-Pair Differential Encoder 140 may supply the differential representation together with the engineered feature differences Δf to the Trust-Inference Engine 160. The Trust-Inference Engine 160 may function as a differential trust model that outputs a change-specific trust effect and feature attribution isolating the effect of the deliberate design intervention. The change-specific trust effect quantifies how the design modification affects the predicted trust-propensity coefficient, enabling reproducible measurement of controlled logo changes rather than comparison of unrelated images.

[0079] FIG. 5 illustrates a counterfactual design-prescription workflow of the Design-Prescription Generator 190 and the Candidate Modification Module 200 according to various embodiments. The workflow generates actionable bounded design modifications rather than general feature-importance rankings.

[0080] The Design-Prescription Generator 190 may select a modifiable feature with an adverse calibrated contribution from the context-calibrated attribution vector. The modifiable feature may be a design parameter that can be adjusted to improve the trust-propensity coefficient, such as stroke weight, inter-character spacing, circularity, edge-density balance, or palette characteristics.

[0081] The Candidate Modification Module 200 may generate bounded candidate parameter changes Δx for the selected modifiable feature. The bounded candidate parameter changes may be constrained within acceptable ranges to preserve brand identity while exploring potential improvements to trust-propensity.

[0082] The Candidate Modification Module 200 may construct candidate logo variations by applying the candidate parameter changes to the original logo representation. Each candidate logo $x + \Delta x$ represents a modified version of the original logo with the proposed parameter change applied.

[0083] Each candidate logo may be re-evaluated against a trust-propensity improvement criterion $\hat{T}(x + \Delta x) - \hat{T}(x) \geq \delta$ and design-preservation constraints. The re-evaluation determines whether the candidate change produces the desired

improvement in trust-propensity while satisfying applicable design-preservation requirements.

[0084] When the improvement satisfies the threshold δ and all constraints are met, the Candidate Modification Module 200 may select the minimum effective change satisfying the criterion. The minimum effective change selection identifies the smallest parameter adjustment that achieves the desired trust-propensity improvement, expressed as $\arg \min \|\Delta x\|$ subject to the improvement criterion and constraints.

[0085] The Reporting and Visualisation Module 210 may output a numerical prescription with before/after score comparison. The numerical prescription specifies the quantitative design modification, such as a specific percentage reduction in stroke weight or a specific increase in inter-character spacing. The before/after score comparison shows the trust-propensity coefficient before and after the proposed modification.

[0086] The Audit and Provenance Store 220 may record an audit trail including the selected modifiable feature, the candidate changes evaluated, the improvement criterion, the constraints applied, and the selected minimum effective change. The audit trail enables reproducibility and verification of the design-prescription process.

[0087] When the improvement does not satisfy the threshold δ or constraints are not met, the Candidate Modification Module 200 may adjust the search range and try the next candidate. The iterative search process continues until an acceptable solution is found or the search space is exhausted. The iterative approach enables the system to explore alternative modifications when initial candidates do not meet the improvement criterion or violate design-preservation constraints.

[0088] The following working examples demonstrate the operation of the Differential Trust Engine 100 in various scenarios. The examples illustrate the adaptive behaviour of the Dynamic Signal Calibration Module 180 and the matched-pair differential analysis capability.

[0089] **Working Example 1: Achromatic Logo Processing**

[0090] In a first working example, a single achromatic logo is processed through the Differential Trust Engine 100. The Image Input and Validation Module 110 receives the achromatic logo image and validates the input. The Image Preprocessing Module 120 performs resizing, normalisation and colour-space conversion. The Multi-Domain Feature Extraction Module 130 extracts chromatic features, geometric features, typographic features and learned image embeddings.

[0091] The Dynamic Signal Calibration Module 180 measures saturation $S = 0.02$, which is below the achromatic threshold $\tau_a = 0.05$. The Achromatic Signal Gate 181 activates and excludes colour-domain attribution values from the prescription vector Φ_{cal} while preserving the raw attribution vector Φ_{raw} . The diagnostic process is redirected to stroke, circularity, symmetry and typography features. The Design-Prescription Generator 190 generates a prescription based on geometric and typographic features, avoiding irrelevant palette advice for the achromatic logo.

15 [0092] **Working Example 2: Matched-Pair Differential Analysis**

[0093] In a second working example, a baseline logo r_0 and a controlled revision r_1 with modified typography are processed through the Differential Trust Engine 100 in matched-pair mode. The Image Input and Validation Module 110 receives and validates both logo images. The Multi-Domain Feature Extraction

20 Module 130 extracts features from both logos.

[0094] The Matched-Pair Differential Encoder 140 computes the embedding difference $\Delta E = E(r_1) - E(r_0)$, the displacement magnitude $d = \|\Delta E\|_2$, and the engineered feature differences Δf capturing the typography modification. The Trust-Inference Engine 160 generates trust-propensity coefficients for both logos and quantifies the change-specific trust effect attributable to the typography modification. The system outputs the trust-propensity change and feature attribution isolating the effect of the deliberate design intervention.

25 [0095] **Working Example 3: Dominant-Colour Logo Processing**

[0096] In a third working example, a dominant-blue logo is processed through the Differential Trust Engine 100. The Multi-Domain Feature Extraction Module

30 130 extracts features including a dominant blue colour channel. The Explainability

Engine 170 generates a raw attribution vector with a raw attribution value of -0.0231 for the dominant blue colour feature.

[0097] The Dynamic Signal Calibration Module 180 measures dominance $D = 0.67$, which exceeds the dominance threshold $\tau_d = 0.40$. The Dominant Signal

5 Contextualiser 182 activates and retains the raw attribution value of -0.0231 while generating a separate contextual preservation/modification score ψ . The contextual score determines that the intentional dominant blue colour should be preserved while another modifiable feature is identified as design friction. The Design-Prescription Generator 190 generates a prescription targeting the identified design
10 friction feature rather than the intentional dominant colour.

[0098] Experimental Validation Results

[0099] The Differential Trust Engine 100 has been integrated and evaluated in a controlled computational laboratory environment. The following table summarises the experimental validation results:

Validation Item	Reported Value	Interpretation
Coefficient of determination (R^2)	0.7344	Approximately 73.44% of target variance explained
Root Mean Squared Error (RMSE)	0.0167	Prediction error on normalised target scale
Processed model-development records	181,866	Combined processed/augmented records
Held-out evaluation proportion	20% (approximately 36,373 records)	Evaluation subset separated from model fitting
General-logo corpus	167,140 images; 2,341 categories	Broad logo representation

Emotion palette resource	745 palettes; 35 psychological dimensions; 3,725 expanded records	Independent psychographic mapping resource
End-to-end processing time	Below 3 seconds per new logo	Laboratory execution time

[0100] The following table summarises the functional validation of adaptive modes:

Test Case	Measured Condition	System Response	Technical Outcome
Achromatic logo	Saturation $S = 0.02$; $\tau_a = 0.05$	Achromatic Gate activated; colour attributions excluded from prescription vector	Diagnostic redirected to stroke, circularity, symmetry and typography; irrelevant palette advice avoided
Dominant-blue logo	Dominance $D = 0.67$; $\tau_d = 0.40$; raw attribution = -0.0231	Raw value retained; separate contextual preservation/modification score generated	Intentional colour preserved while another modifiable feature identified as design friction

[0101] The following table summarises the integrated subsystem status:

Subsystem	Laboratory Status
Image ingestion and preprocessing	Integrated
Feature extraction	Integrated
Matched-pair displacement computation	Integrated

Emotion-manifold lookup	Integrated
Trust inference and ensemble	Integrated
Raw explainability generation	Integrated
Achromatic and dominant-signal mode selection	Integrated
Context-calibrated attribution	Integrated
Feature-specific prescription and reporting	Integrated

[0102] It should be further appreciated that reference throughout the specification to features, operations, or components using singular terms should not be construed as excluding a plurality thereof unless the context expressly indicates otherwise. Similarly, features described in the context of grouped or listed elements (e.g., A, B, and C) should be understood to encompass individual, multiple, or all combinations of said elements. Where ranges are given, all intermediate values and subranges are understood to be disclosed as if specifically recited. The scope of the invention should therefore be construed as inclusive of all such combinations and logical extensions as would be appreciated by a person of ordinary skill in the relevant technical field.

[0103] In another embodiment, the above disclosure is a description of the invention and is not intended to limit the scope of the invention. Other variations and modifications of the above-described embodiment shall be apparent to those skilled in the art and are intended to fall within the scope of the invention as defined in the following claims.

Dated This 24th day of July 2026



(Kuldeep Singh)

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Authorized Agent for the Applicant,
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WE CLAIM:

1. A method for context-adaptive brand-logo trust-propensity inference and design prescription, the method comprising:

receiving and validating a logo image at an Image Input and Validation
5 Module (110);

preprocessing the logo image at an Image Preprocessing Module (120);

extracting multi-domain features including chromatic features, geometric features, typographic features and a learned image embedding from the logo image at a Multi-Domain Feature Extraction Module (130);

10 mapping a representative colour vector to neighbouring emotion-labelled palette vectors at an Emotion-Manifold Mapping Module (150);

generating a trust-propensity coefficient from the extracted multi-domain features at a Trust-Inference Engine (160);

generating a raw attribution vector at an Explainability Engine (170);

15 classifying a chromatic operating condition and generating a context-calibrated attribution vector while preserving the raw attribution vector at a Dynamic Signal Calibration Module (180);

selecting a modifiable feature and generating a quantitative design prescription at a Design-Prescription Generator (190); and

20 outputting a trust-propensity score, explanations and design prescription at a Reporting and Visualisation Module (210).

2. The method of claim 1, wherein the Dynamic Signal Calibration Module (180) comprises an Achromatic Signal Gate (181) that excludes colour-domain attribution values from a prescription pathway when measured saturation
25 is below a configurable achromatic threshold.

3. The method of claim 2, wherein the Dynamic Signal Calibration Module (180) comprises a Dominant Signal Contextualiser (182) that retains the raw attribution vector and generates a contextual preservation/modification score when a dominance ratio exceeds a configurable dominance threshold.

30 4. The method of claim 1, further comprising storing both the raw attribution vector and the context-calibrated attribution vector along with

thresholds, model version and selected analysis mode at an Audit and Provenance Store (220).

5. The method of claim 1, further comprising:

receiving a baseline logo and a controlled revision at the Image Input and
5 Validation Module (110); and
computing an embedding difference and an embedding displacement
between the baseline logo and the controlled revision at a Matched-Pair Differential
Encoder (140).

6. The method of claim 5, wherein the Matched-Pair Differential
10 Encoder (140) retains engineered feature differences to support feature-specific
interpretation of a design intervention.

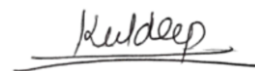
7. The method of claim 1, wherein the Emotion-Manifold Mapping
Module (150) implements nearest-neighbour search through a KD-tree indexing
structure using a corpus of palettes tagged across psychological dimensions.

15 8. The method of claim 1, further comprising combining outputs from
a differential model and a general-logo model using configurable weights at an
Ensemble Corroboration Module (165).

9. The method of claim 1, wherein the Design-Prescription Generator
(190) selects a modifiable feature with an adverse calibrated contribution and
20 computes a bounded change comprising at least one of reduction of stroke weight,
increase of inter-character spacing, adjustment of circularity, reduction of edge-
density imbalance, or constrained palette adjustment.

10. The method of claim 1, further comprising constructing and re-
evaluating bounded logo variations at a Candidate Modification Module (200) by
25 evaluating candidate changes and selecting a minimum change satisfying an
improvement criterion and applicable design-preservation constraints.

Dated This 24th day of July 2026



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ABSTRACT

**CONTEXT-ADAPTIVE EXPLAINABLE ARTIFICIAL INTELLIGENCE
SYSTEM FOR BRAND-LOGO TRUST-PROPENSITY INFERENCE AND
AUTOMATED DESIGN PRESCRIPTION**

- 5 A method for context-adaptive brand-logo trust-propensity inference includes extracting multi-domain features (130) from a logo image, mapping a colour vector to emotion-labelled palettes (150), generating a trust-propensity coefficient (160), generating raw feature attributions (170), classifying a chromatic operating condition and producing a context-calibrated attribution vector while preserving the
- 10 raw attribution vector at a Dynamic Signal Calibration Module (180), generating a quantitative design prescription (190), and outputting the trust-propensity score and design prescription (210).

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